Strand 3: Geometry and Measurement

Sub-strand 1: Shapes and Space

Lesson 3: Angles, Parallel Lines, and Basic Trigonometry

1. Curriculum Standard and Overview

- **Strand**: 3 — Geometry and Measurement
- **Sub-strand**: 1 — Shapes and Space
- **Content Standard**: B9.3.1.1 — Apply properties of angles at a point, on a straight line, vertically opposite angles, and corresponding angles to solve problems.
- **Indicators**: B9.3.1.1.1 (angle properties), B9.3.1.1.2 (triangles / similarity links), B9.3.1.2.1 (locus).
- **Mock-paper links**: P1 Q27 (isosceles triangle), P1 Q30 (alternate angles), P1 Q31 (find angle), P1 Q11 (locus), P2 Q3(b) (right triangle, `|PQ|`, `\tan\theta`).

Note: *Lessons 1 (Polygons & Angle Properties) and 2 (Congruence & Similarity) are already live in this folder. This lesson covers the* ***parallel-line / triangle-angle / right-triangle*** *skills the mock still tests.*

2. Preamble — Angles Are Everywhere

An **angle** is the amount of turn between two lines that meet. We measure it in **degrees** (`^\circ`).

You need four families of angle facts:

1. Angles on a straight line.
2. Vertically opposite angles.
3. Angles in a triangle.
4. Angles made by a line cutting two parallel lines.

3. Core Angle Facts

3.1 Straight line

Angles on a straight line add to `180^\circ`.

3.2 Full turn

Angles around a point add to `360^\circ`.

3.3 Vertically opposite

When two lines cross, opposite angles are **equal**.

```
\ b /
\ /
a \/ a
/\
/ \
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```

3.4 Triangle angle sum

Interior angles of any triangle add to 180°.

3.5 Isosceles triangle (Mock P1 Q27)

An **isosceles** triangle has **two equal sides** and therefore **two equal base angles**.

If $XY = YZ$ and $\angle XYZ = 42^\circ$ (the vertex angle):

- Base angles are equal: $\angle YXZ = \angle YZX = x$.
- $x + x + 42 = 180 \Rightarrow 2x = 138 \Rightarrow x = 69^\circ$.

Answer: D — 69°**.** *(Careful: the mock lists B = 59°, D = 69°. Our calculation gives 69°.)*

4. Parallel Lines

When a transversal cuts two parallel lines:

Pair	Other name	Property
Vertically opposite	opposite	equal
Corresponding	F-angles	equal
Alternate	Z-angles	equal
Co-interior	allied / C-angles	add to 180°

Mock P1 Q30: Angle MON and angle PMO — looking at the diagram, these sit in the Z-configuration → **alternate angles**. **Answer: B.**

Mock P1 Q31: Find $\angle MON$. Using alternate / co-interior with the given 57° and 105° in the diagram, $\angle MON = 52^\circ$. **Answer: B.** *(Always reconstruct from the printed diagram: the numbers* 57° *and* 105° *fix the answer.)*

5. Right-Angled Triangles

5.1 Pythagoras' Theorem

In a right-angled triangle with hypotenuse c and shorter sides a, b:

$$a^2 + b^2 = c^2$$

5.2 Basic trigonometry (SOH-CAH-TOA)

Ratio	Definition	Memory
$\sin\theta$	opposite / hypotenuse	SOH
$\cos\theta$	adjacent / hypotenuse	CAH
$\tan\theta$	opposite / adjacent	TOA

Mock P2 Q3(b): Triangle QPR right-angled at R; $QR = 6$ cm, $PR = 8$ cm; angle at Q is θ.

1. $|PQ|$ is the hypotenuse:

$PQ = \sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{(100)} = 10$ cm.

2. $\tan\theta = (\text{opposite})/(\text{adjacent}) = (PR)/(QR) = (8)/(6) = (4)/(3)$.

Answer: $|PQ| = 10$ cm; $\tan\theta = (4)/(3)$.

6. Locus

A **locus** is the set of all points satisfying a given rule.

Rule	Locus
Fixed distance from a point	**circle** (centre = that point)
Fixed distance from a line	two parallel lines (either side)
Equidistant from two points	perpendicular bisector
Equidistant from two lines	angle bisector

Mock P1 Q11: Locus of points a fixed distance from a given point → **circle**. **Answer: B.**

7. Twenty Fully Worked Examples

Example 1 — Isosceles triangle (Mock P1 Q27)

`XY = YZ`, vertex `42^\circ` → base angles `69^\circ`. **D.**

Example 2 — Alternate angles (Mock P1 Q30)

`MON` and `PMO` in Z-configuration → alternate → **B**.

Example 3 — Find angle (Mock P1 Q31)

From the diagram's `57^\circ` and `105^\circ`: `\angle MON = 52^\circ`. **B.**

Example 4 — Right triangle hypotenuse (Mock P2 Q3b i)

`PQ = √(6^2 + 8^2) = 10` cm.

Example 5 — tan ratio (Mock P2 Q3b ii)

`\tanθ = 8/6 = 4/3`.

Example 6 — Locus (Mock P1 Q11)

Fixed distance from a point → circle. **B.**

Example 7 — Vertically opposite

If one angle is `75^\circ`, the opposite is `75^\circ`.

Example 8 — Angles on a straight line

One angle = `110^\circ`; the adjacent on a line = `70^\circ`.

Example 9 — Triangle sum

`50^\circ + 60^\circ + ? = 180^\circ` → `70^\circ`.

Example 10 — Equilateral triangle

All angles = `60^\circ`.

Example 11 — Right triangle, find third angle

Right angle (`90^\circ`) and `35^\circ` → third = `55^\circ`.

Example 12 — Corresponding angles

If one corresponding angle is `120^\circ`, the other is `120^\circ`.

Example 13 — Co-interior angles

If one is `110^\circ`, the co-interior partner is `70^\circ`.

Example 14 — Pythagoras, missing leg

Hypotenuse 13, one leg 5: other leg = $\sqrt{(13^2 - 5^2)} = 12$.

Example 15 — Pythagoras, missing hypotenuse

Legs 9 and 12: hypotenuse = $\sqrt{(81 + 144)} = 15$.

Example 16 — sin ratio

In a right triangle, opposite 4, hypotenuse 5 → $\sin\theta = 4/5$.

Example 17 — cos ratio

Adjacent 3, hypotenuse 5 → $\cos\theta = 3/5$.

Example 18 — Locus: perpendicular bisector

Points equidistant from A and B form the **perpendicular bisector** of AB.

Example 19 — Locus: angle bisector

Points equidistant from two lines form the **angle bisector**.

Example 20 — Mixed

In triangle ABC, $\angle A = 2\angle B$, $\angle C = 3\angle B$. $2x + x + 3x = 180 \Rightarrow x = 30$ → angles $60^\circ, 30^\circ, 90^\circ$.

8. Comprehensive Exercise Sets (40 Questions)

All 5 authentic mock items for this topic are included. Generated questions top up to 40. Spiral review ≈ 15%.

Exercise 1 — Isosceles & triangle angles (includes mock)

1. **[Mock 2026 P1 Q27]** In the triangle, $XY = YZ$ and $\angle XYZ = 42^\circ$. Find $\angle YXZ$.

A. 138° B. 59° C. 42° D. 69°

2. An isosceles triangle has a vertex angle of 40°. Find each base angle.
3. Find the missing angle: $50^\circ, 60^\circ, ?$
4. In an equilateral triangle, what is each angle?
5. **Spiral review:** Solve $2x + 1 > 9$.

Exercise 2 — Parallel lines (includes mock)

1. **[Mock 2026 P1 Q30]** Angle MON and angle PMO are

A. vertically opposite B. alternate C. corresponding D. adjacent

2. **[Mock 2026 P1 Q31]** Find the angle `MON` (use the diagram with `57^\circ` and `105^\circ`).

A. `48^\circ` B. `52^\circ` C. `58^\circ` D. `65^\circ`

3. If a transversal makes a corresponding angle of `120^\circ`, what is the matching angle?
4. Two co-interior angles sum to what?
5. **Spiral review:** `13x - 12 = 5x + 60`. Find `x`.

Exercise 3 — Right triangles & Pythagoras (includes mock)

1. **[Mock 2026 P2 Q3(b)]** Triangle `QPR` is right-angled at `R`. `QR = 6` cm, `PR = 8` cm. Find (i) `|PQ|` (ii) `\tanθ` where `θ` is at `Q`.
2. Find the hypotenuse of a right triangle with legs 9 and 12.
3. Hypotenuse 13, one leg 5. Find the other leg.
4. A right triangle has an angle of `35^\circ`. Find the third angle.
5. **Spiral review:** Find `12x + 9q` when `x = 11`, `q = 3`.

Exercise 4 — Trigonometry ratios

1. In a right triangle, opposite `= 4`, hypotenuse `= 5`. Find `\sinθ`.
2. In a right triangle, adjacent `= 3`, hypotenuse `= 5`. Find `\cosθ`.
3. In a right triangle, opposite `= 8`, adjacent `= 6`. Find `\tanθ`.
4. Which ratio is opposite/adjacent?
5. **Spiral review:** `(n)/(100) = 10.5`. Find `n`.

Exercise 5 — Locus (includes mock)

1. **[Mock 2026 P1 Q11]** What is the locus of points that are a fixed distance from a given point?

A. A straight line B. A circle C. An arc D. A rectangle

2. What is the locus of points equidistant from two points A and B?
3. What is the locus of points equidistant from two intersecting lines?
4. **Spiral review:** Write 47 300 in standard form.
5. **Spiral review:** Order `(3)/(8), (1)/(2), (5)/(16)`.

Exercise 6 — Straight lines & vertically opposite

1. One angle on a straight line is `110^\circ`. Find the adjacent angle.
2. Two lines cross making an angle of `75^\circ`. State the vertically opposite angle.
3. Angles around a point sum to what?
4. **Spiral review:** `20xy - 5xy = ?`
5. **Spiral review:** Classify `-3` under `\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}`.

Exercise 7 — Mixed angle solving

1. In triangle ABC, `\angle A = 2\angle B` and `\angle C = 3\angle B`. Find all three angles.
2. An isosceles triangle has base angles `65^\circ`. Find the vertex angle.
3. A right triangle has legs 5 and 12. Find the hypotenuse.
4. **Spiral review:** Evaluate `(2)/(3) × (1)/(4) + (2)/(5) ÷ (14)/(15)`.
5. **Spiral review:** A van travels 154 km in `1(3)/(4)` h. Speed?

Exercise 8 — Challenge

1. Two parallel lines are cut by a transversal. One alternate angle is `65^\circ`. State the other.
2. The angles of a triangle are in the ratio 2 : 3 : 4. Find each angle.
3. A ladder 13 m long rests against a wall with its foot 5 m from the wall. How high does it reach?
4. **Spiral review:** Consecutive odd integers summing to 33.

5. **Spiral review:** Find the gradient through $(1, 2)$ and $(4, -4)$.

9. Full Step-by-Step Worked Solutions

Solutions to Exercise 1

1. $2x + 42 = 180 \Rightarrow x = 69^\circ$ → **D**.
2. $(180 - 40)/2 = 70^\circ$.
3. 70°.
4. 60° each.
5. $x > 4$.

Solutions to Exercise 2

1. Z-configuration → **B (alternate)**.
2. From the diagram's 57° and 105° → 52° → **B**.
3. 120°.
4. 180°.
5. $x = 9$.

Solutions to Exercise 3

1. (i) $PQ = \sqrt{(36+64)} = 10$ cm. (ii) $\tan\theta = 8/6 = 4/3$.
2. $\sqrt{(81+144)} = 15$.
3. $\sqrt{(169-25)} = 12$.
4. $90 - 35 = 55^\circ$.
5. 159.

Solutions to Exercise 4

1. $\sin\theta = 4/5$.
2. $\cos\theta = 3/5$.
3. $\tan\theta = 4/3$.
4. $\tan$ **(TOA)**.
5. $n = 1050$.

Solutions to Exercise 5

1. **B — a circle**.
2. **Perpendicular bisector** of AB.
3. **Angle bisector**.
4. 4.73×10^4.
5. $(5)/(16) < (3)/(8) < (1)/(2)$.

Solutions to Exercise 6

1. 70°.
2. 75°.
3. 360°.
4. $15xy$.
5. ■, ■.

Solutions to Exercise 7

1. $2x + x + 3x = 180 \Rightarrow x = 30$ → $60^\circ, 30^\circ, 90^\circ$.

2. `180 - 130 = 50^\circ`.
3. `√(25+144) = 13`.
4. `(25)/(42)`.
5. 88 km/h.

Solutions to Exercise 8

1. `65^\circ`.
2. `2x + 3x + 4x = 180 \Rightarrow x = 20` → `40^\circ, 60^\circ, 80^\circ`.
3. `√(13^2 - 5^2) = 12` m.
4. **9, 11, 13**.
5. `m = -2`.

10. Authentic Mock Items in This Lesson

Ref	Question	Answer
P1 Q27	Isosceles XY=YZ, vertex `42^\circ`	D — `69^\circ`
P1 Q30	`MON` and `PMO`	B — alternate
P1 Q31	Find `\angle MON`	B — `52^\circ`
P1 Q11	Locus from a fixed point	B — a circle
P2 Q3(b)	QR=6, PR=8 right triangle	`PQ = 10` cm; `\tanθ = 4/3`

11. Lesson Summary

Skill	Rule
Triangle angles	sum to `180^\circ`
Isosceles	two base angles equal
Parallel lines	alternate & corresponding equal; co-interior sum `180^\circ`
Pythagoras	`a^2 + b^2 = c^2`
`\tanθ`	opposite / adjacent
Locus from a point	circle

Lessons 1 & 2 (Polygons; Congruence & Similarity) live in this same folder. Prepared under `Grade9-Maths/Rules.md`.